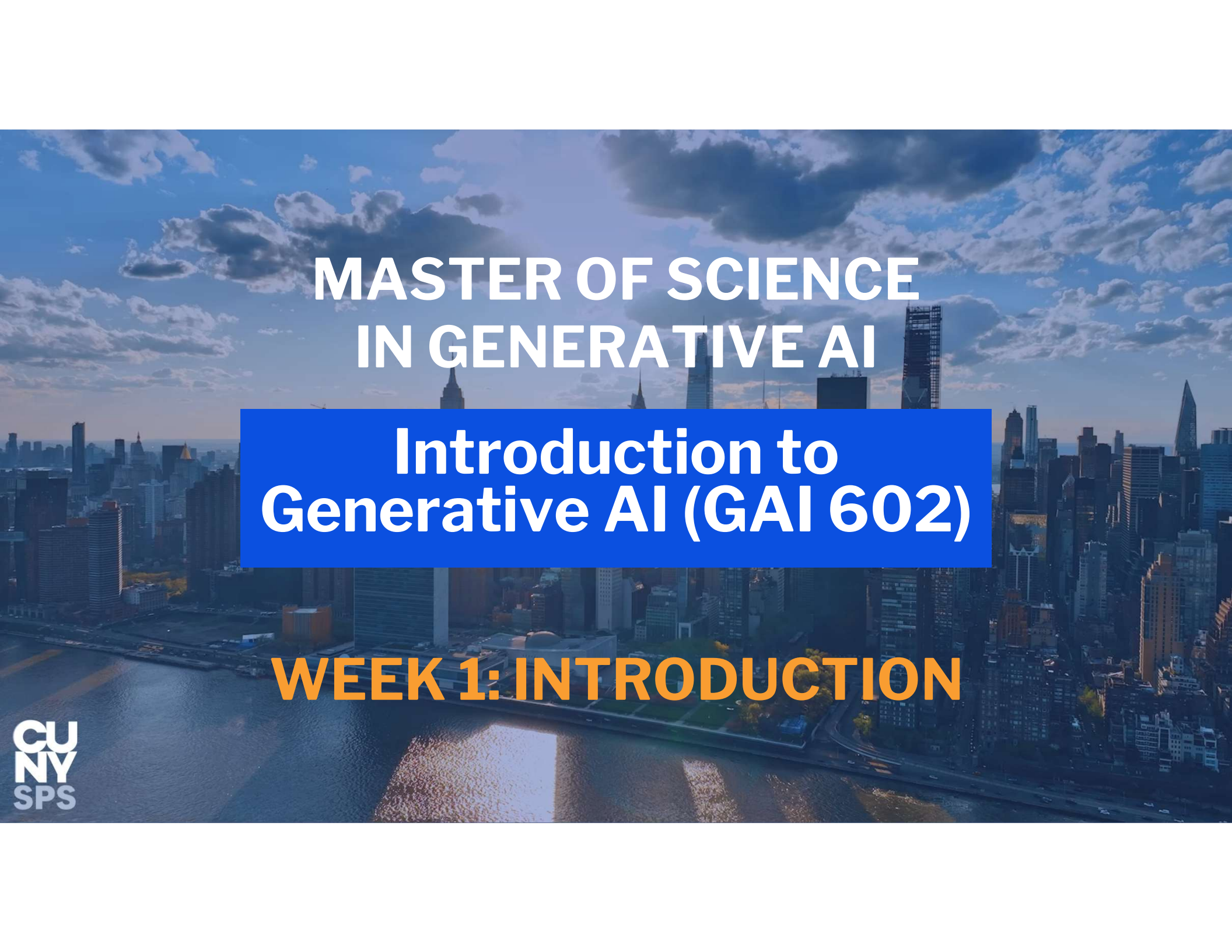




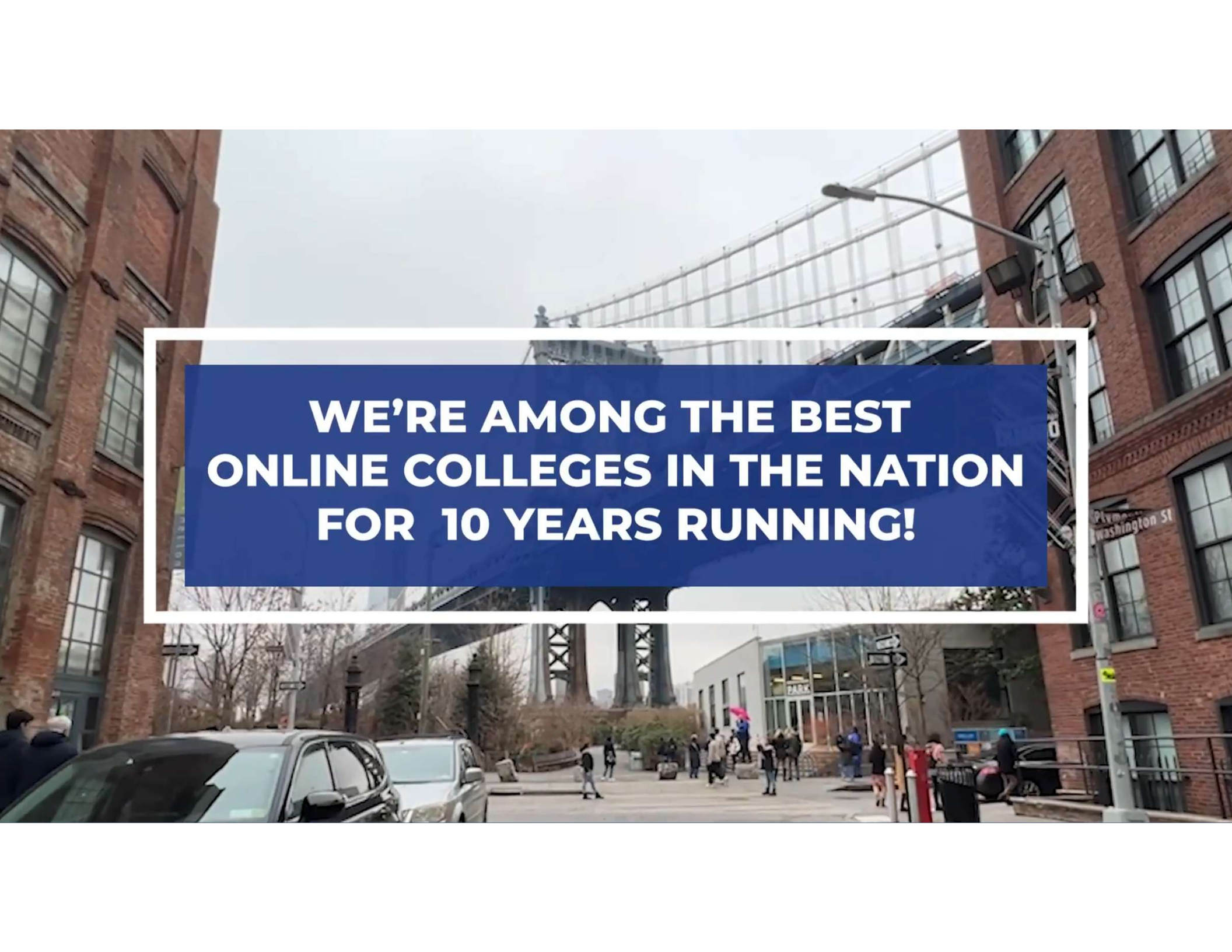
MASTER OF SCIENCE IN GENERATIVE AI

Introduction to Generative AI (GAI 602)

WEEK 1: INTRODUCTION

A photograph of a busy New York City street corner. In the foreground, a cyclist wearing a bright yellow-green jacket and a black helmet is riding a bicycle across the frame. To the left, several yellow taxis are visible, some stopped at a crosswalk. Pedestrians are walking on the sidewalk. The background features a large, ornate building with arched windows and a street sign that reads "West 34th St" and "Sixth Ave". A blue text overlay with white text is centered in the image.

**HIGHEST RANKED
ONLINE COLLEGE IN
NEW YORK CITY!**

A street scene in Pittsburgh, featuring brick buildings on either side and the iconic steel structure of the Fort Pitt Bridge in the background. A large blue rectangular overlay with white text is centered in the upper half of the image. The text reads: "WE'RE AMONG THE BEST ONLINE COLLEGES IN THE NATION FOR 10 YEARS RUNNING!". The street below shows pedestrians and cars, with a street sign for "Plymouth Washington St" visible on the right.

**WE'RE AMONG THE BEST
ONLINE COLLEGES IN THE NATION
FOR 10 YEARS RUNNING!**



Top Data Science Programs Integrating AI, ML, Data Engineering, and Generative AI

Leading universities are integrating AI, machine learning, data engineering, MLOps, and generative AI into data science programs, preparing professionals for end-to-end AI development and enterprise deployment.

Written By : Poulami Saha

Reviewed By : Achu Krishnan

Published: 24th Jun, 2026 at 11:45 PM

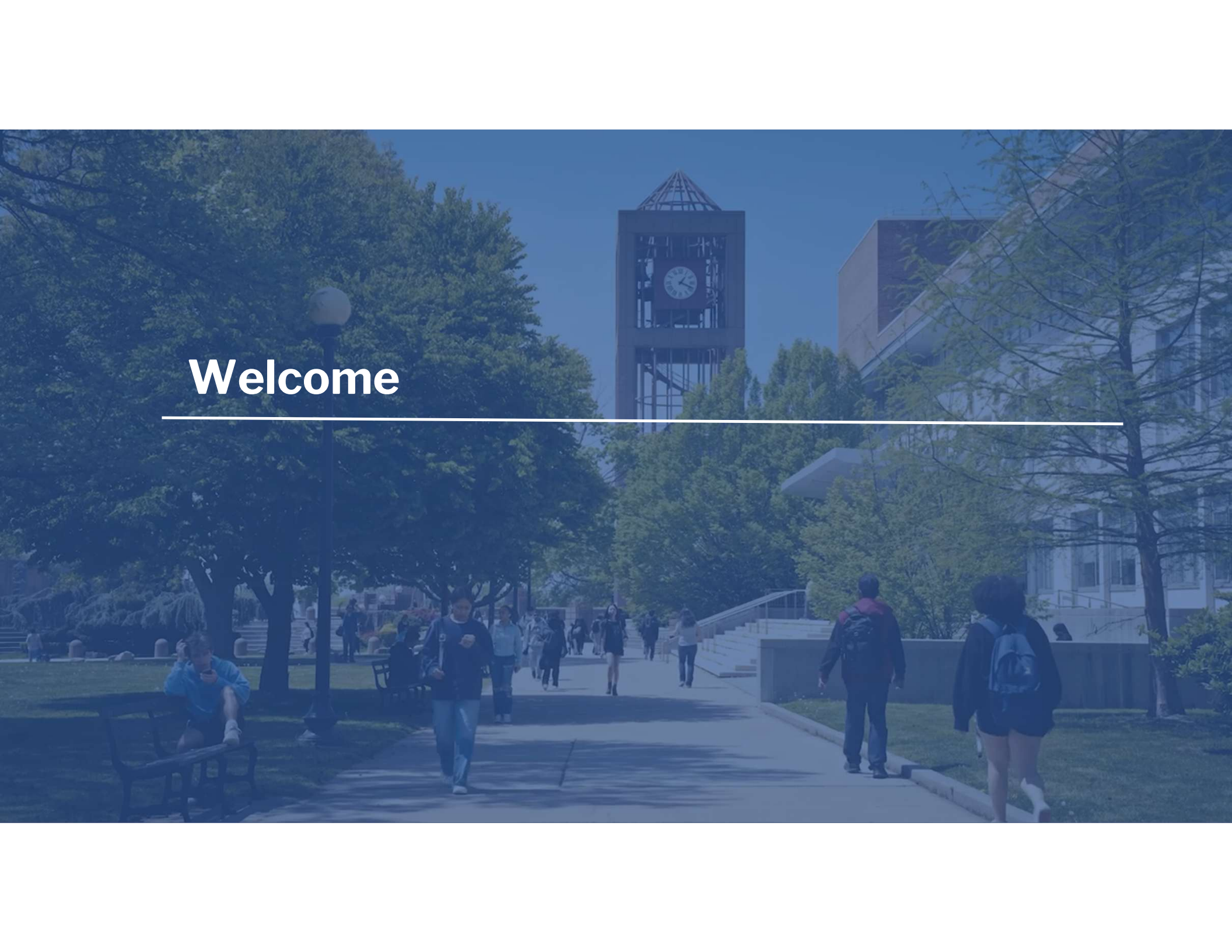
Top 7 Data Science Programs Integrating AI, ML, Data Engineering and Generative AI

Institution	AI & ML	Data Engineering	Generative AI	Industry Focus	Best For
MIT	Excellent	Strong	Strong	Research & Innovation	AI Researchers
CMU	Excellent	Strong	Strong	Applied AI	ML Engineers
Stanford	Excellent	Strong	Strong	Startups & Innovation	AI Entrepreneurs
UC Berkeley	Strong	Excellent	Strong	Enterprise Analytics	Data Scientists
TU Delft	Strong	Strong	Moderate	Responsible AI	AI System Designers
IIT Jammu	Strong	Strong	Excellent	Enterprise AI	Working Professionals
CUNY SPS	Moderate	Moderate	Excellent	Generative AI	GenAI Specialists



Introduction to Generative AI (GAI602)

Joe Sabelja

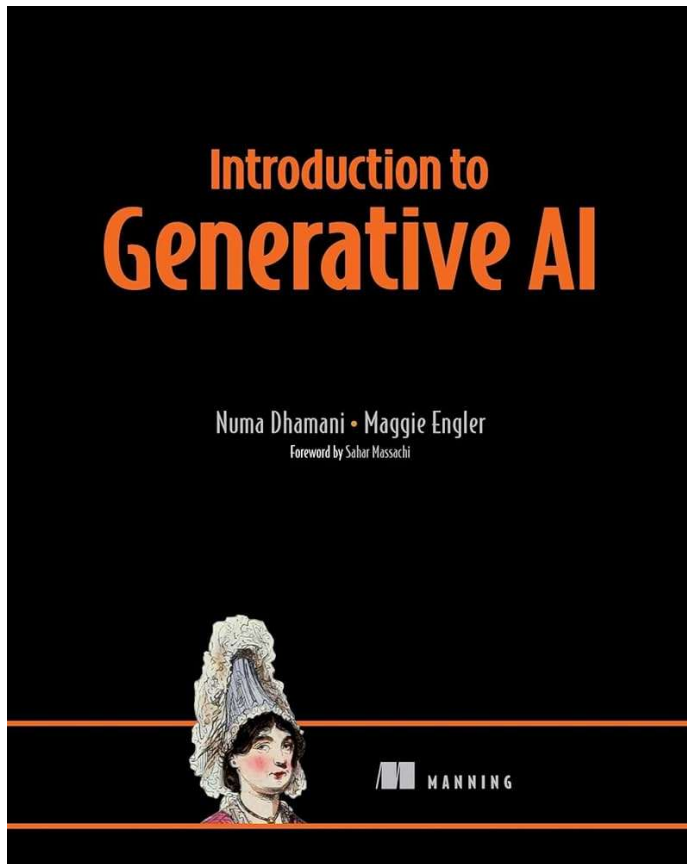


Welcome

Syllabus

1. Overview of the LLM Ecosystem
2. Tokenization
3. Sequence-to-Sequence Approaches
4. Transformers
5. Attention Mechanisms
6. Training Large Language Models (LLMs)
7. Prompt Engineering
8. Semantic Search and Embeddings
9. Fine-Tuning & Retrieval-Augmented Generation (RAG)
10. Agents and Plugins
11. Inference & Quantization
12. LLM MLOps (LMOps)
13. Ethics and Responsible AI
14. Scaling Laws and Model Evaluation

Textbook



Introduction to Generative AI

by Numa Dhamani (co-authored with Maggie Engler)

February 2024

ISBN-10: 1633437191

ISBN-13: 978-1633437197

Schedule

2026	September					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
31 Week 1	01	02	03 Class 7PM	04	05	06 Introductions
07 Labor Day	08 Week 2	09	10 Class 7PM	11 Rosh Hashana	12	13 Quiz 1*
14 Week 3	15	16	17 Class 7PM	18	19	20 Asst 1
21 Yom Kippur	22 Week 4	23	24 Class 7PM	25	26	27 Quiz 2
28 Week 5	29	30	01	02	03	04

2026	October					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
28 Week 5	29	30	01 Class 7PM	02	03	04 Asst 2
05 Week 6	06	07	08 Class 7PM	09	10	11 Quiz 3
12 Columbus Day	13 Week 7	14	15 Class 7PM	16	17	18 Mid-term
19 Week 8	20	21	22 Class 7PM	23	24	25 Feedback
26 Week 9	27	28	29 Class 7PM	30	31	01

2026	November					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
26	27	28	29 Class 7PM	30	31	01 Asst 3
02 Week 10	03	04	05 Class 7PM	06	07	08 Quiz 4
09 Week 11	10	11	12 Class 7PM	13	14	15 Asst 4
16 Week 12	17	18	19 Class 7PM	20	21	22 Quiz 5
23	24	25	26	27	28	29

Thanksgiving: CUNY SPS offices closed, no classes scheduled*

2026	December					
MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
30 Week 13	01	02	03 Class 7PM	04	05	06 Asst 5
07 Week 14	08	09	10 Class 7PM	11	12	13 Final Exam
14	15	16	17	18	19	20
21 Fall Semester Ends	22	23	24	25	26	27
28	29	30	31	01	02	03

Final Examinations

Holiday observed - CUNY SPS is closed

Final Grades due

Holiday observed - CUNY SPS is closed

* Work due during a holiday can be submitted the next day.

Class Policies

1. Please leave your camera on

- See: https://www.cuny.edu/wp-content/uploads/sites/4/page-assets/academics/faculty-affairs/Camera-Use-Guidance-for-Online-and-Hybrid-Courses_FINAL-JUNE-20-2024.pdf

2. Review the Course Policies on Brightspace

- Respect, Online Etiquette and Anti-Harassment
- Lateness Policy

3. Read the guidelines on Generative AI & Academic Integrity on Brightspace

- Provide Snapshots with answers
- Do not use Generative AI for quizzes and exams
- You (the human) manage AI, not the other way around
- Don't cheat yourself

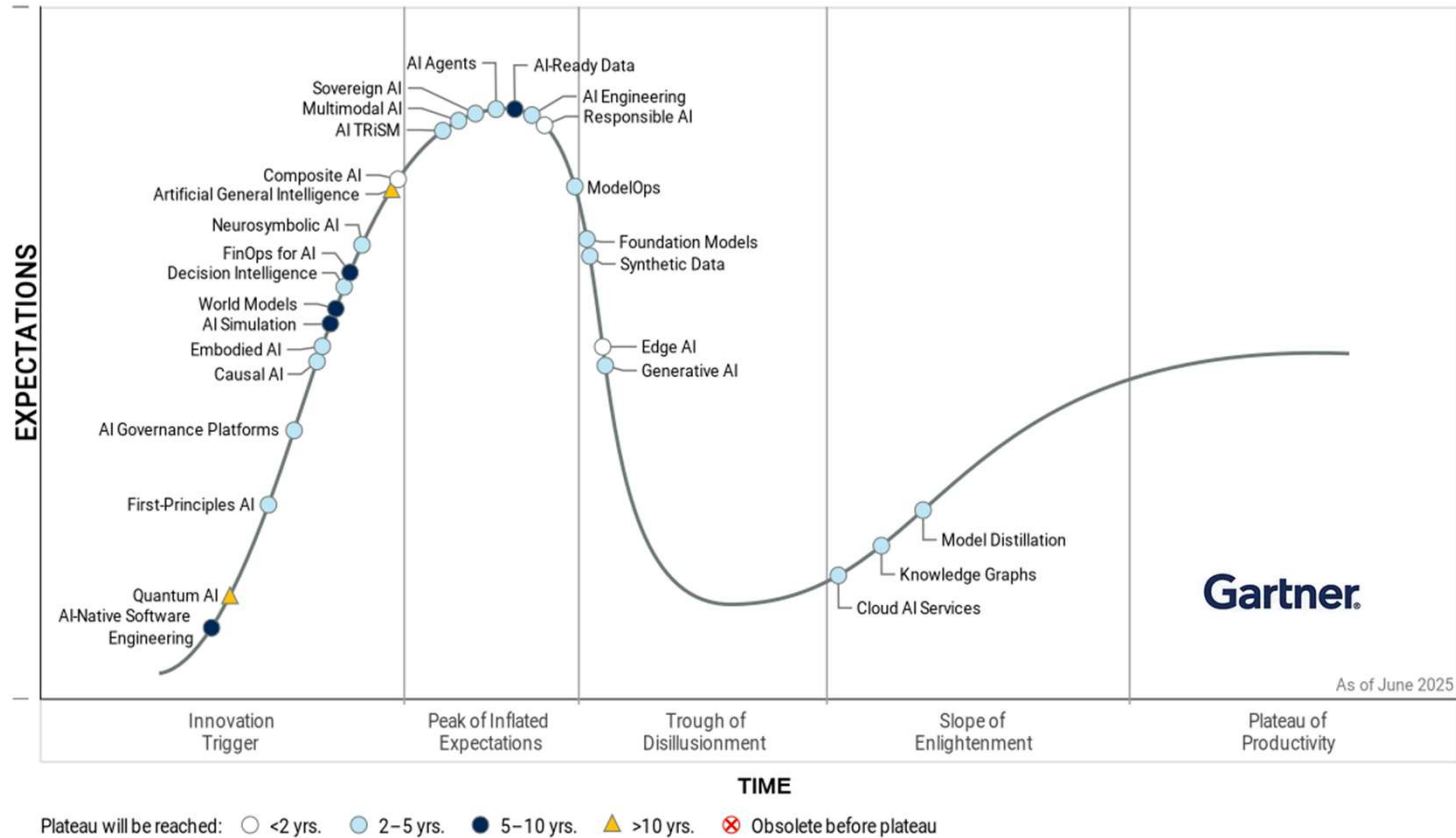
Grading

Component	Number	Total Value
Quizzes	5	20%
Assignments	4	20%
Exam: Mid-term	1	30%
Exam: End of semester	1	30%
		100%

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground is filled with lush green trees and a wide, paved pedestrian walkway. Several students are visible: some are walking along the path, while one student is sitting on a wooden bench on the left. The overall atmosphere is academic and serene.

Imperative of Generative AI

Generative AI is an overnight sensation



Source: Gartner

Sample use-cases

Research and development	Manufacturing	Sales and marketing	Supply chain management	Retailer and distribution	Enabling areas	Finance	Sales and marketing	Customer service	Operations
Autonomous product innovation	Autonomous quality control	Multi-channel campaign orchestrator	Demand forecasting agent	Shelf stocking assistant	Autonomous finance analyst	Under-writing recommendation agent	Cross-sell recommendation agent	Autonomous case resolution agent	Loan processing agent
Smart experimentation	Predictive maintenance agent	Dynamic pricing agent	Inventory balancer	Planogram compliance checker	HR talent scout	Risk reasoning assistant	Autonomous promotions campaigns	KYC/Onboarding compliance agent	Claims adjudication agent
Patent landscape navigator	Production scheduling optimiser	Sentiment-driven brand manager	Logistics route optimiser	Retail forecast collaboration agent	Compliance and risk monitoring agent	Dispute resolution agent	AI sales advisor agent	Voice of customer agent	Fraud detection agent
Consumer sentiment synthesiser	Energy efficiency advisor	Autonomous content creator	Supplier risk monitor	Autonomous retail order management	Cybersecurity sentinel	Expense anomaly agent	Dynamic pricing agent	Grievance escalation agent	Payment reconciliation agent
Sustainability optimiser	Waste reduction agent	Retailer-specific offer optimiser	Autonomous procurement agent	Autonomous sales beat planning	ESG reporting agent	Regulatory reporting agent	Competitor intelligence agent	Proactive service agent	
AI-driven competitive analysis	AI-driven process optimisation	Autonomous customer segmentation	Disruption detection and mitigation agent	Geo-targeted promotion agent	Legal contract review agent				Branch/Channel performance agent

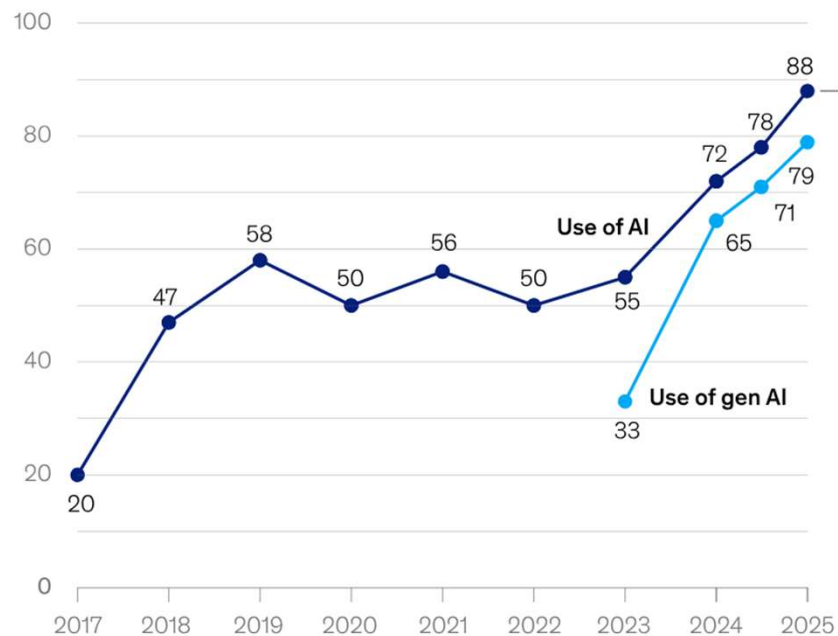
Source: Deloitte Insights, 2025

AI has become a critical need for firms

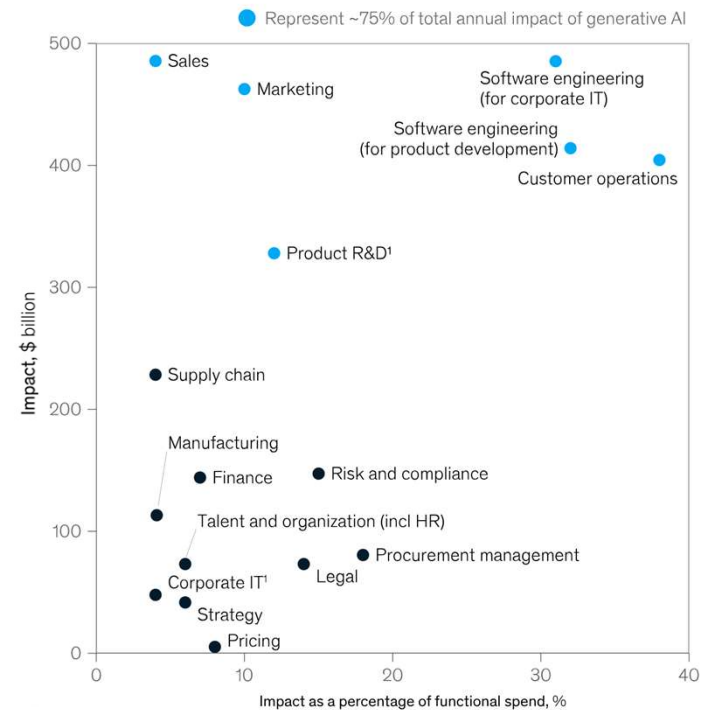
Reported use of AI in at least one business function continues to increase.

Use of AI by respondents' organizations, % of respondents

Organizations that use AI in at least 1 business function¹

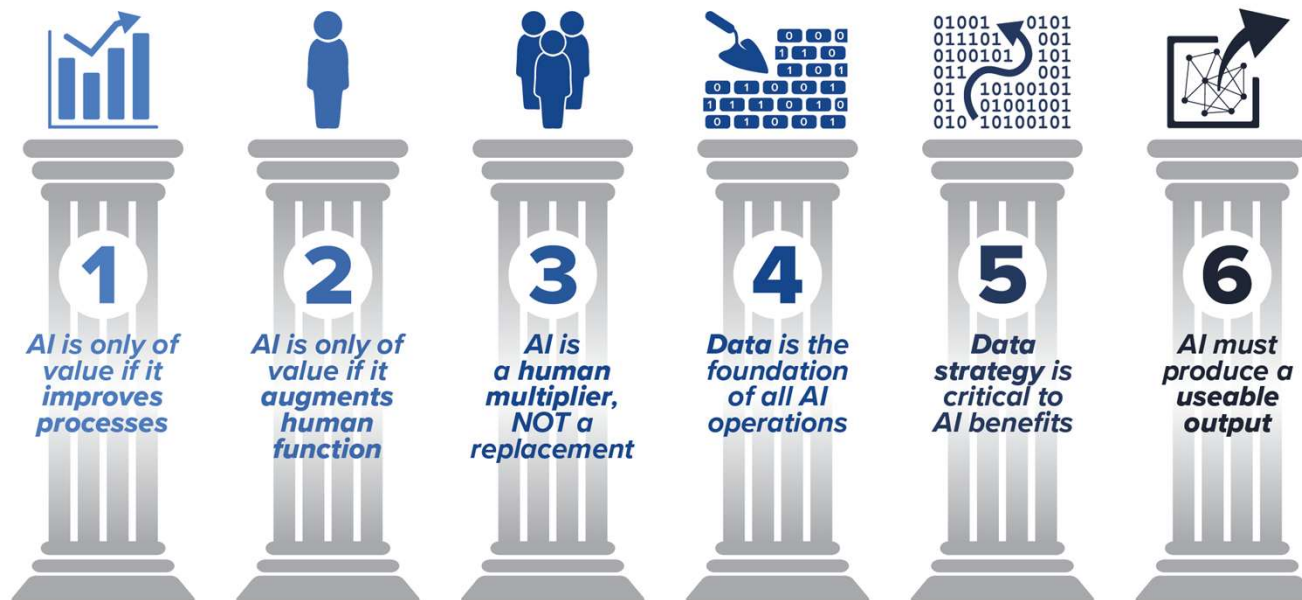


Using generative AI in just a few functions could drive most of the technology's impact across potential corporate use cases.



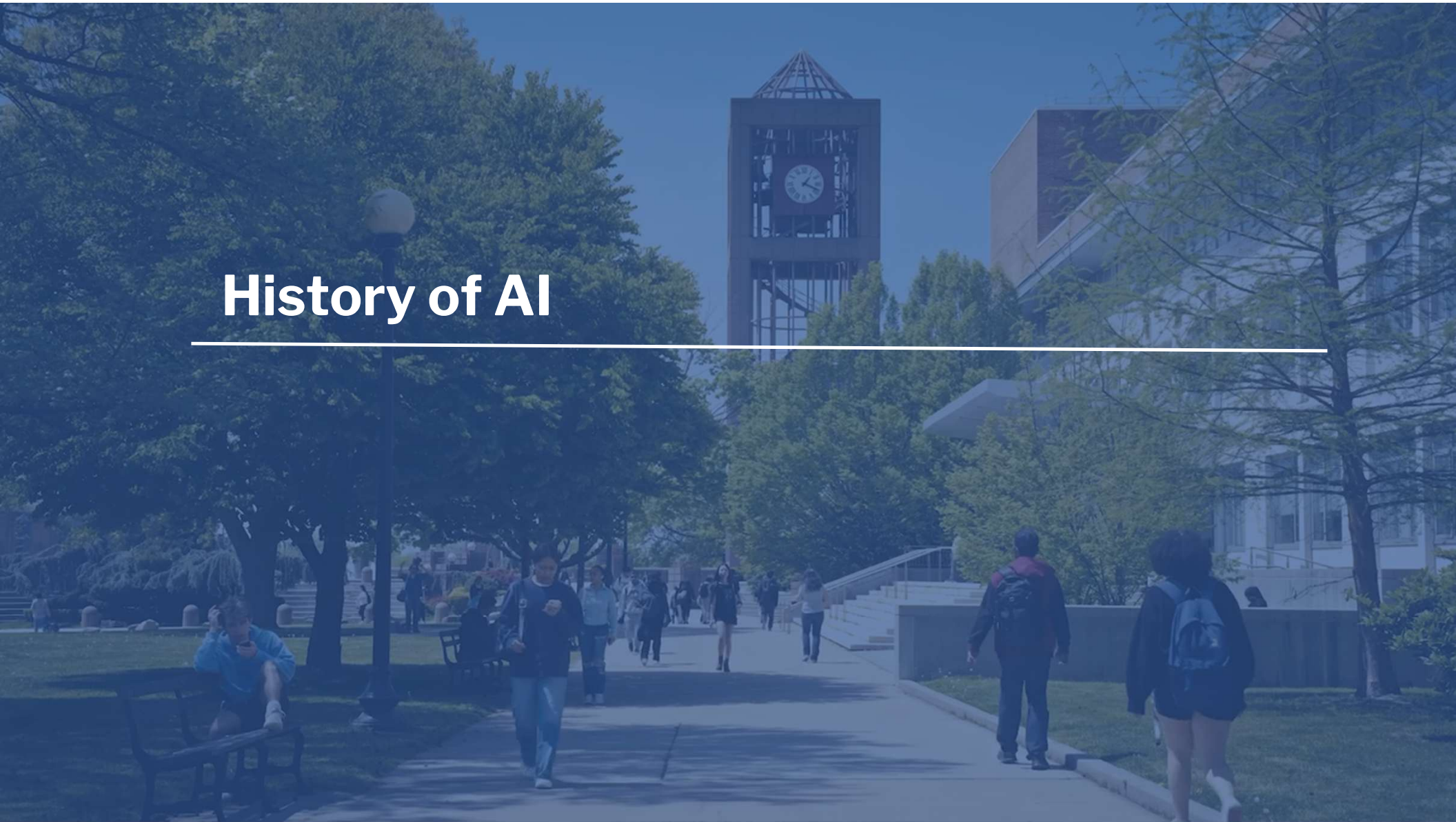
Source: McKinsey

Choosing the right use-cases



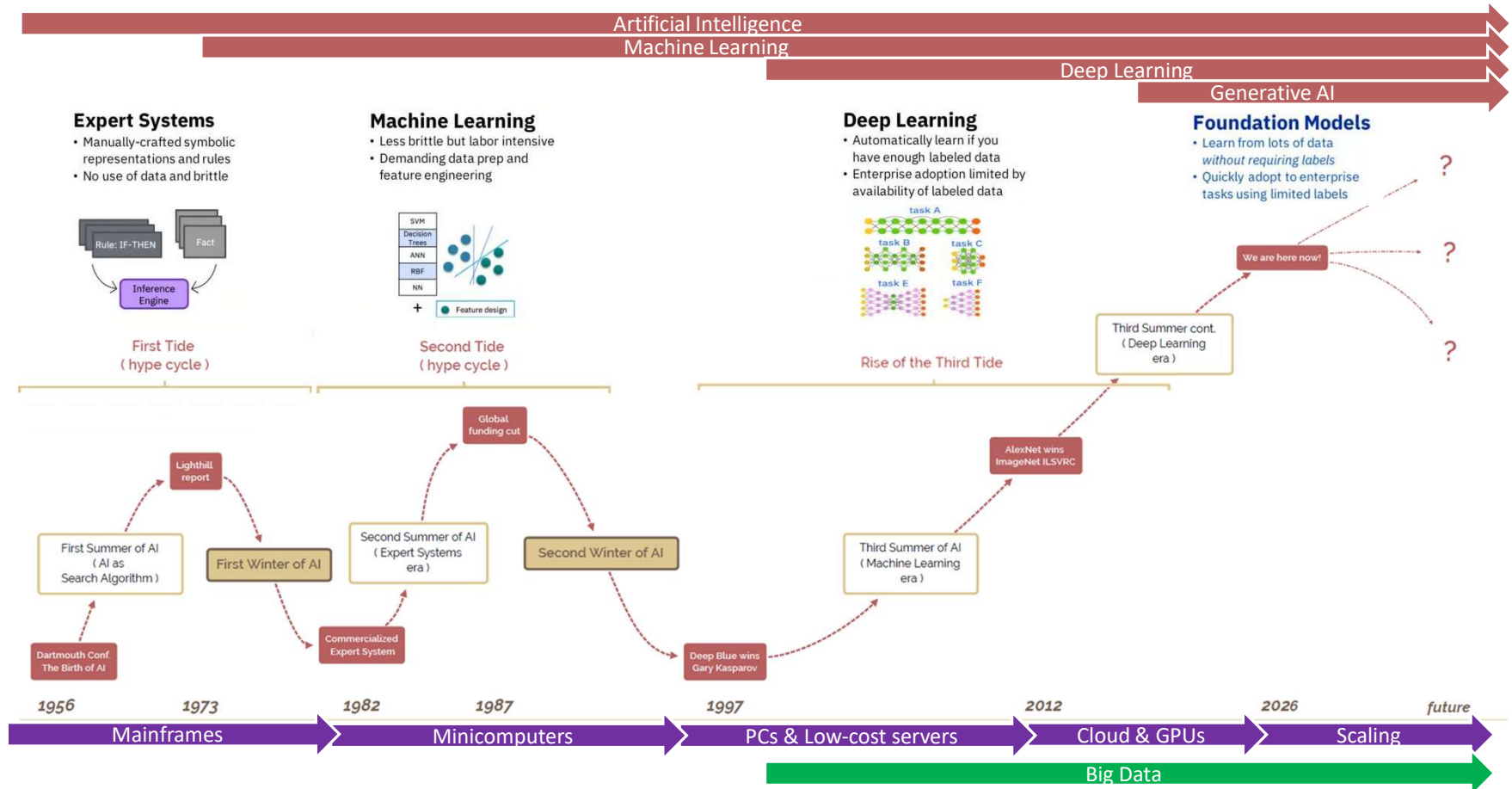
Source: Synergetics

History of AI



[illegible]

AI is fed by technological breakthroughs



Source: A brief history of AI, Amirhosein Toosi, et al, 2022

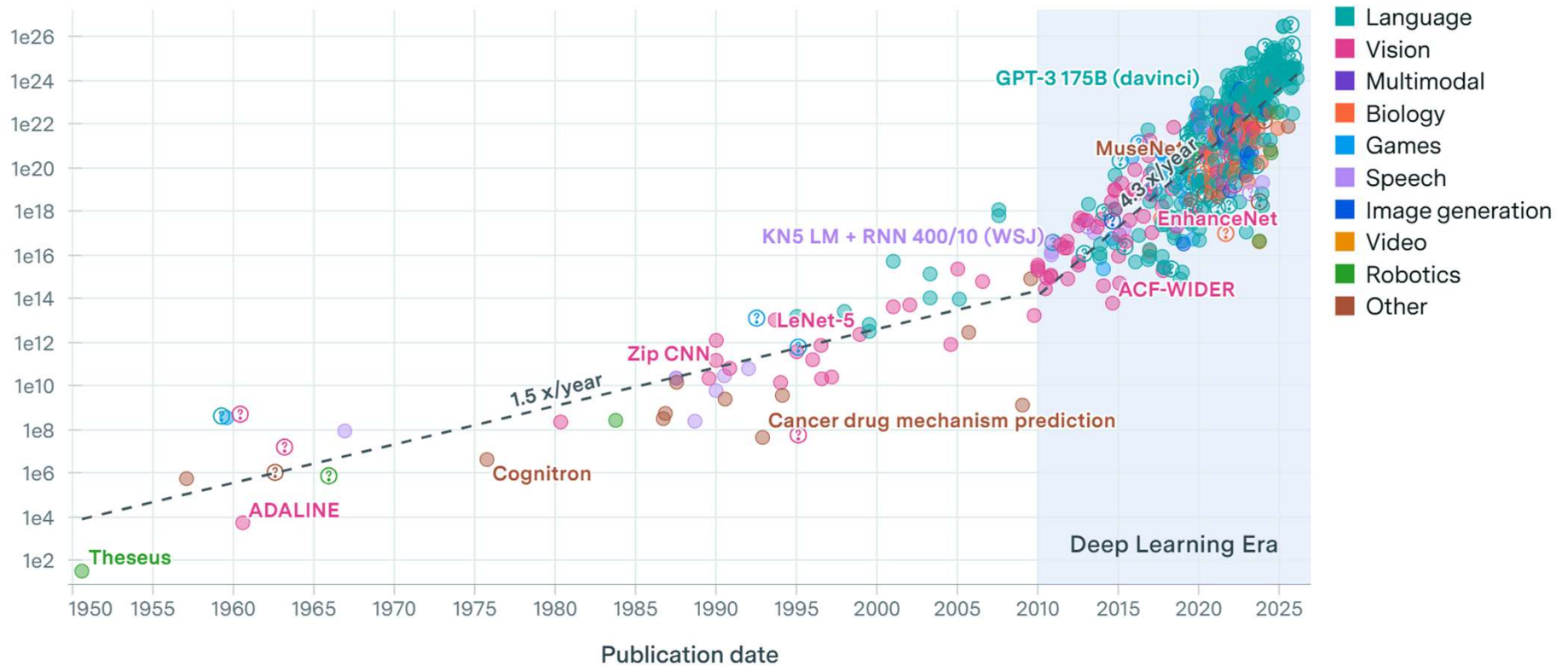
Growth is accelerating – due to GPUs

Notable AI models

EPOCH AI

Training compute (FLOP)

② : Speculative data 540 Results Domain



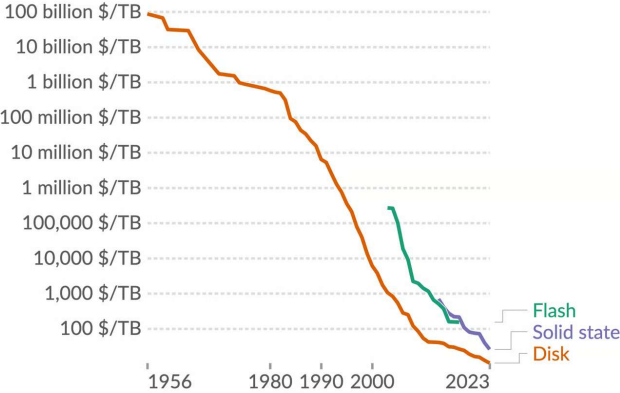
Source: epoch.ai

AI is becoming cheaper

Historical price of computer storage

Our World
in Data

Expressed in US dollars per terabyte (TB), adjusted for inflation. "Disk" refers to magnetic storage, "flash" to memory used for rapid data access and rewriting, and "solid state" to solid-state drives (SSDs).

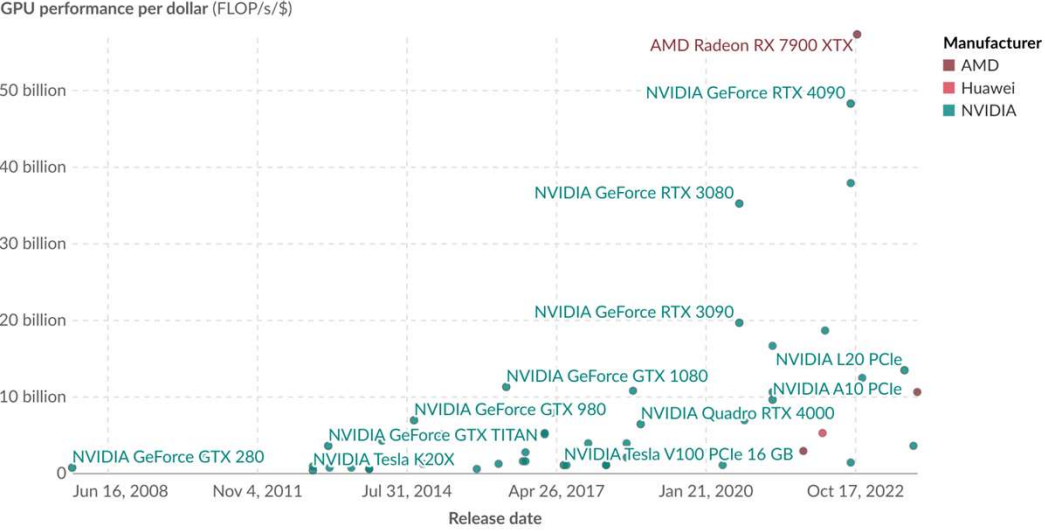


Data source: John C. McCallum (2023) CC BY

GPU computational performance per dollar

Our World
in Data

Hardware computational performance shown in floating-point operations¹ per second (FLOP/s) per US dollar, adjusted for inflation.



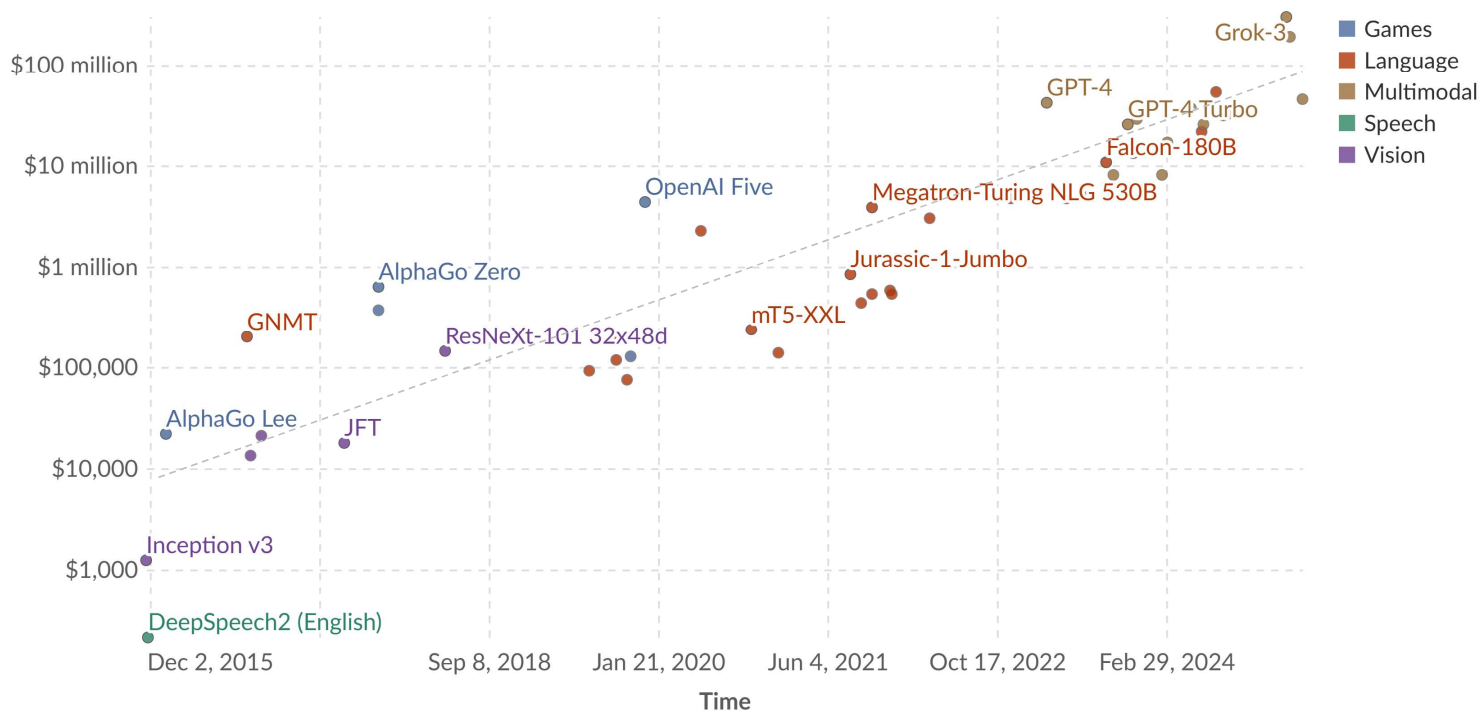
Source: Our world in data

Generative AI models are getting bigger

Hardware and energy cost to train notable AI systems

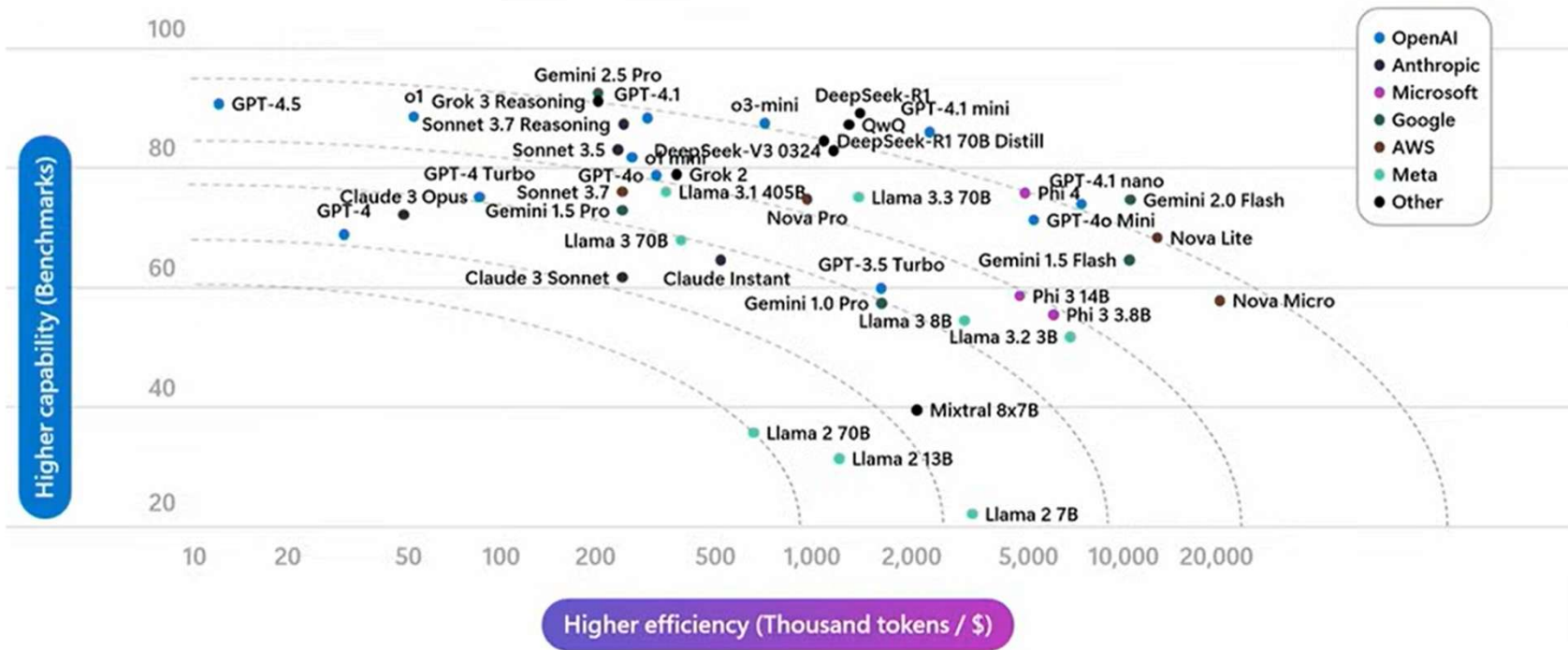
This data is expressed in US dollars, adjusted for inflation.

Cost (constant 2023 US\$; plotted on a logarithmic axis)



Source: Our world in data

An Explosion of Foundation Models

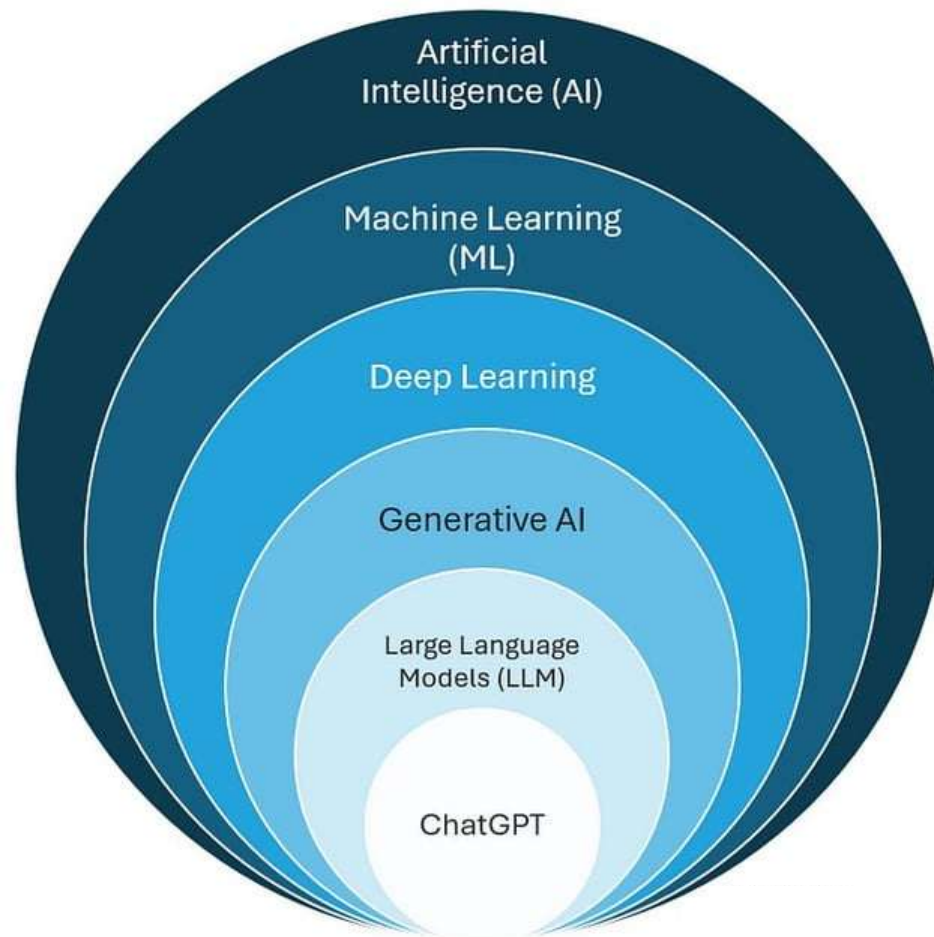


Source: Microsoft

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed top and a visible clock face stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground is filled with lush green trees and a wide, paved pedestrian walkway. Several students are walking along the path; some are carrying backpacks, and one student is sitting on a bench on the left. The overall atmosphere is that of a busy academic environment.

What is Artificial Intelligence?

What is Artificial Intelligence?



What is Artificial Intelligence?

1950's	 Artificial intelligence (AI) <i>Human intelligence exhibited by machines</i>		
	1980's	 Machine learning <i>AI systems that learn from historical data</i>	
		2010's	 Deep learning <i>Machine learning models that mimic human brain function</i>
			2020's

What is intelligence?

"A fundamental problem in artificial intelligence is that nobody really knows what intelligence is."

-Legg & Hutter (2007)

"... the ability to solve hard problems." M. Minsky

"The ability to use memory, knowledge, experience, understanding, reasoning, imagination and judgement in order to solve problems and adapt to new situations." AllWords Dictionary, 2006

"The capacity to acquire and apply knowledge." The American Heritage Dictionary, fourth edition, 2000

"Intelligence is not a single, unitary ability, but rather a composite of several functions. The term denotes that combination of abilities required for survival and advancement within a particular culture." A. Anastasi

"Intelligence is the ability for an information processing system to adapt to its environment with insufficient knowledge and resources." P. Wang

"... the ability to plan and structure one's behavior with an end in view." J. P. Das

"... doing well at a broad range of tasks is an empirical definition of 'intelligence' " H. Masum

"... the ability of a system to act appropriately in an uncertain environment, where appropriate action is that which increases the probability of success, and success is the achievement of behavioral subgoals that support the system's ultimate goal." J. S. Albu

"Intelligence measures an agent's ability to achieve goals in a wide range of environments."

S. Legg and M. Hutter

Source: A Collection of Definitions of Intelligence, Shane Legg, Marcus Hutter, 2007 arXiv:0706.3639

LLM Ecosystem



What is Data Science?

“At a high level, data science is a set of fundamental principles that guide the extraction of knowledge from data.”

Principles can be statistical, computational, algorithmic, visual, etc.

Source: NVIDIA DLI Accelerated Data Science

What is Artificial Intelligence?

1950's	 Artificial intelligence (AI) <i>Human intelligence exhibited by machines</i>		
	1980's	 Machine learning <i>AI systems that learn from historical data</i>	
	2010's	 Deep learning <i>Machine learning models that mimic human brain function</i>	
	2020's	 Generative AI (Gen AI) <i>Deep learning models (foundation models) that create original content</i>	

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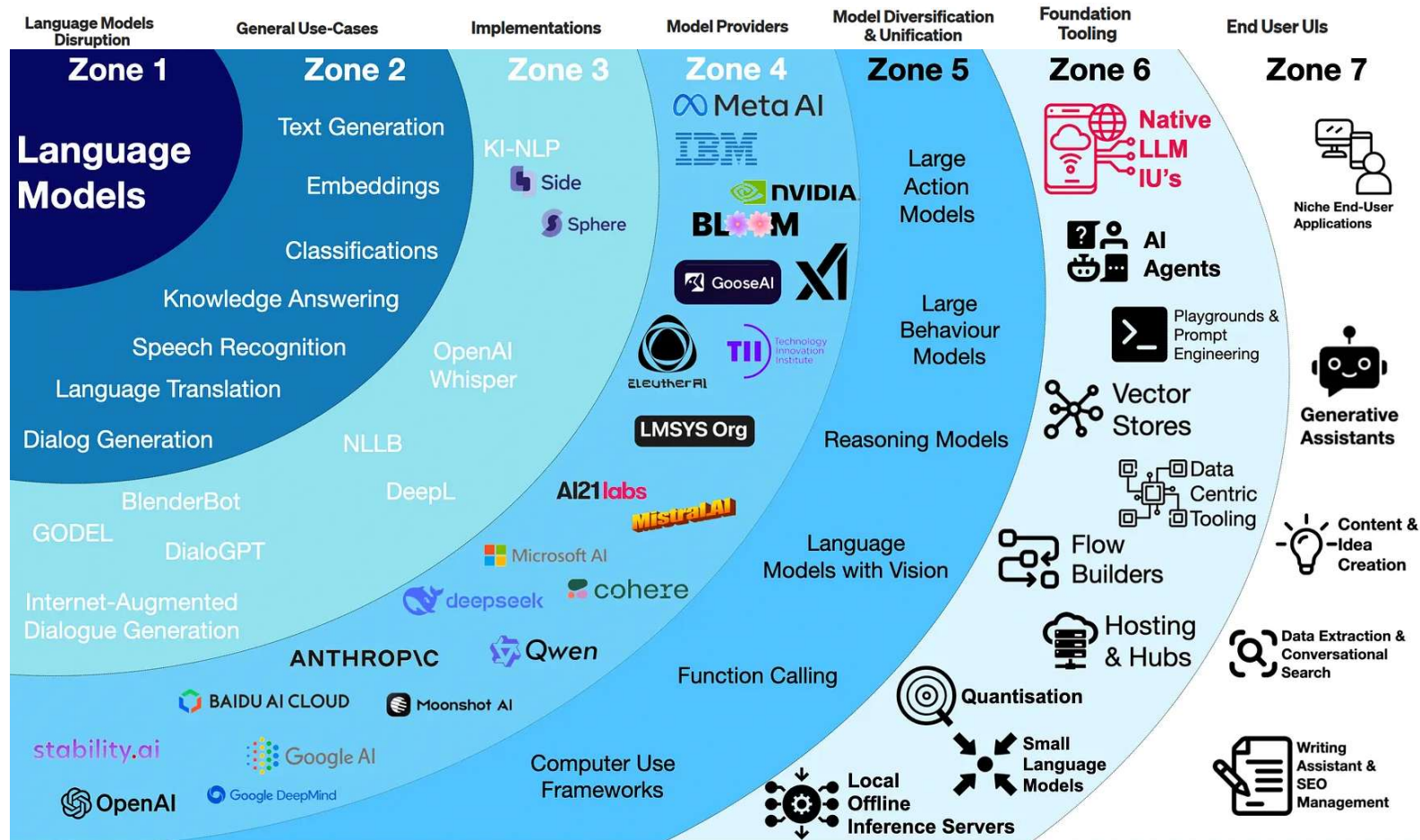
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LLM Ecosystem



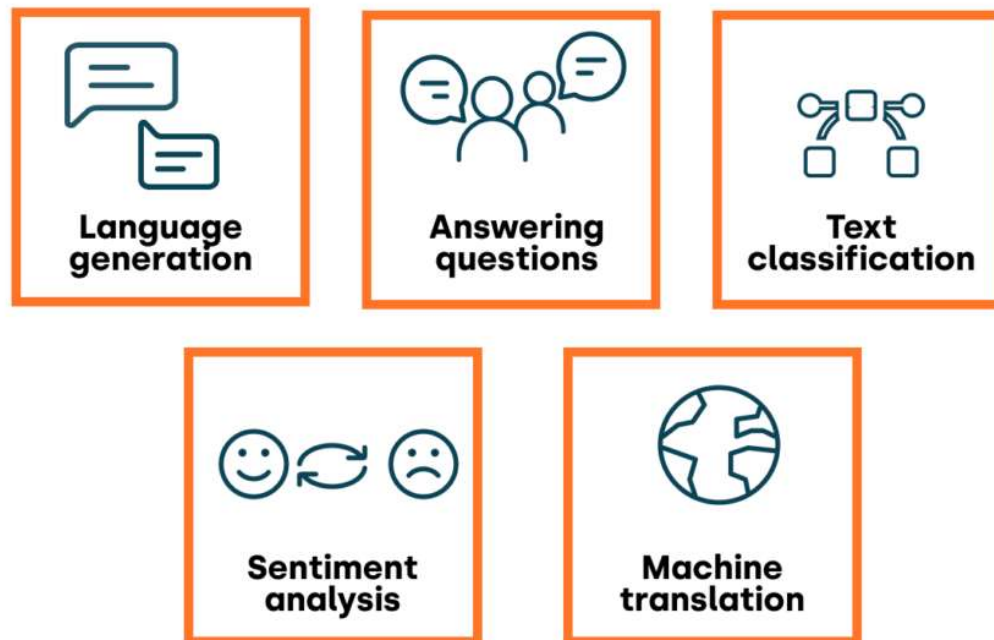
LLM Landscape



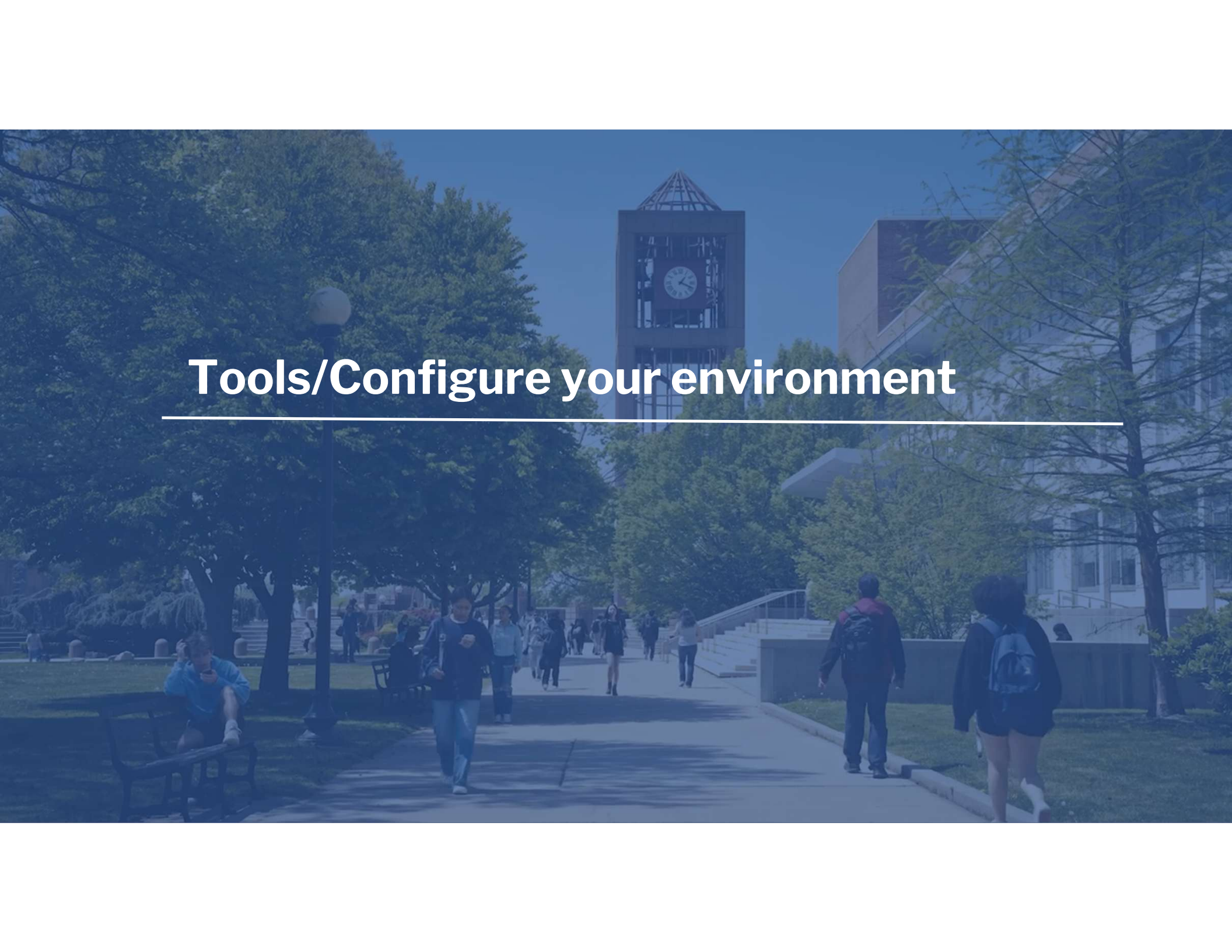
Source: Cobus Greyling

Natural Language Processing

NLP is the process through which **AI is taught to understand the rules and syntax of language**, programmed to develop complex algorithms to represent those rules, and then made to **use those algorithms to carry out specific tasks** like these.



Source: Zapier

A blue-tinted photograph of a university campus. In the background, a tall clock tower with a glass-enclosed clock face and a pyramidal roof stands prominently. To its right is a modern, multi-story building with a grid-like facade. The foreground shows a wide, paved pedestrian walkway lined with mature trees. Several students are walking along the path; some are carrying backpacks. On the left, a student is sitting on a wooden bench. The overall atmosphere is that of a busy academic environment.

Tools/Configure your environment

Register for free tools

1. Colab

- Go to <https://colab.research.google.com/signup>
- Create a Google account (if you don't have one)
- Sign up for **Colab Pro for Education** (free for students)

2. GitHub Copilot

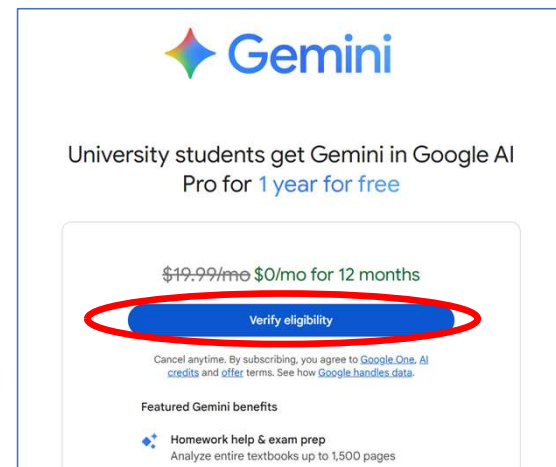
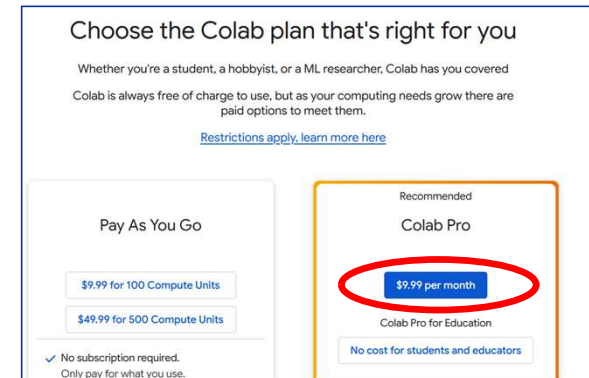
- Go to <https://github.com/education>
- Click on **Join GitHub Education**
- Create a GitHub account (if you don't have one)
- Click on Start an Application to get access

3. Google Gemini

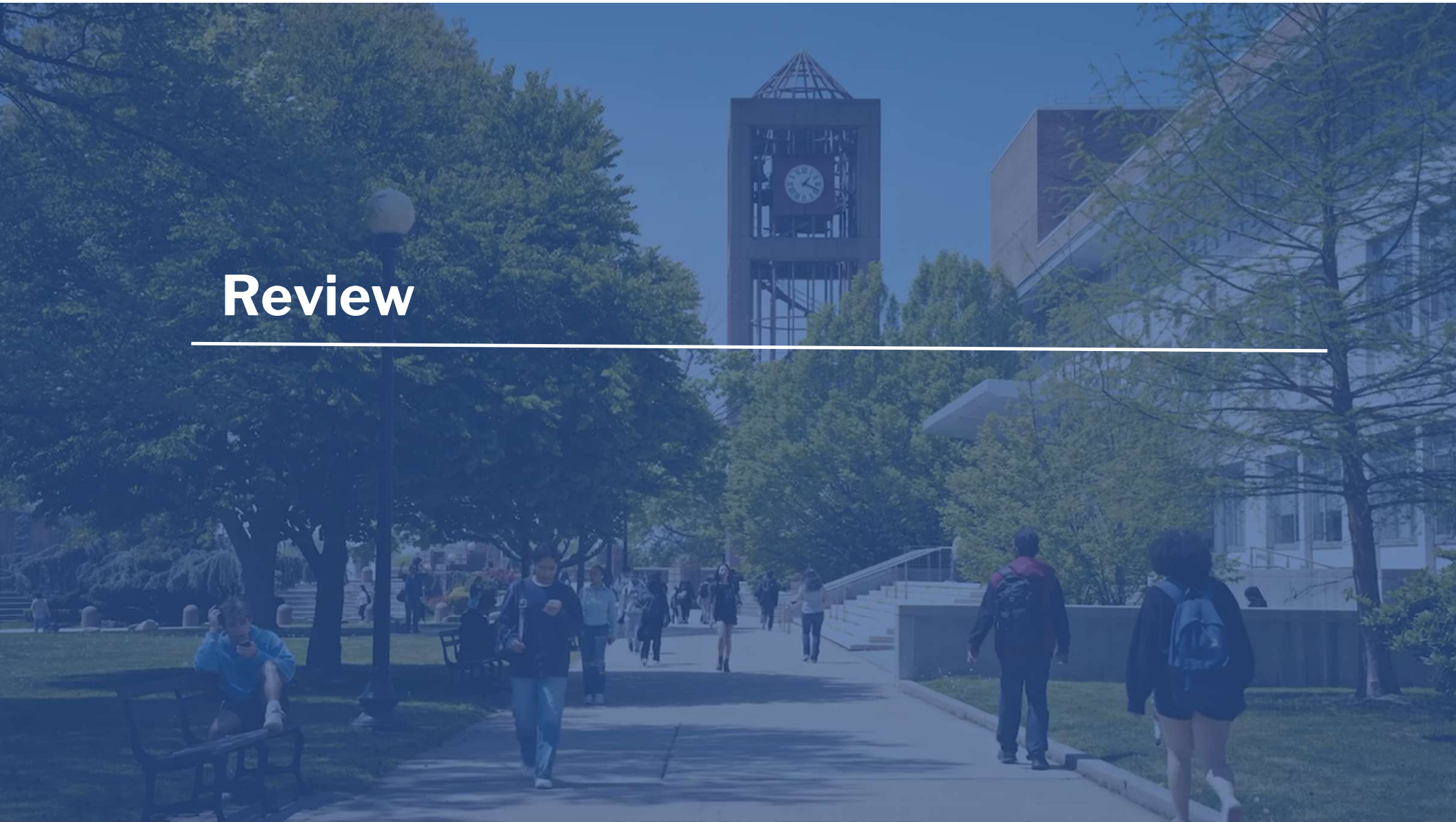
- Go to <https://gemini.google/us/students>
- Fill out the verification form

4. Microsoft Copilot 365

- Free with you Office 365 access (via CUNY)



Review



This week we covered

1. Supervised & Unsupervised Learning

Define the basic types of machine learning (supervised, unsupervised) using examples from real-world applications.

2. Business Role

Explain the role of machine learning in modern business contexts and decision-making.

3. Tooling

Set up and configure Python (or R) environments for machine learning tasks.

4. ML Algorithms

Identify appropriate ML techniques for structured and unstructured business data.

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's stone towers and suspension cables are prominent. The water of the Hudson River is visible in the foreground. The sky is clear and blue.

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